

GNU-Plot-Lab-Notebook

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Section 1

Question 1

Plot the following functions for $-10 \leq x \leq 10$ on the same graph using GNUPLOT. Set 1000 samples. Take the range of y -values from -4 to $+4$. Mark each curve with different line styles and line widths. Label x -axis and y -axis as 'x-values' and 'y-values' respectively. Set the title of the plot as 'Function plots'. Save the plot in a suitable format.

$$f_1(x) = \coth(x) - \frac{1}{x}, \quad f_2(x) = e^{-\sin(x)}, \quad f_3(x) = x|x|$$

Sol: The GNU-Plot code for the solution of this problem is given below.

```
1 reset
2 set terminal svg background "white"
3 set output "Lab-files/Output/1/Q_xxi_ou.svg"
4
5 set title "Function plots"
6 set grid
7 set xzeroaxis linestyle 1 lc "black"
8 set yzeroaxis linestyle 1 lc "black"
9 set xrange [-10:10]
10 set yrange [-4:4]
11 set xlabel "x-values"
12 set ylabel "y-values"
13 set samples 1000
14
15 f1(x)=(1/(tanh(x)))-(1/x)
16 f2(x)=exp(-sin(x))
17 f3(x)=x*abs(x)
18
19 plot f1(x) lw 1 lc "#5f0f40" dt 1, f2(x) lw 2 lc "#fb8b24" dt 2, f3(x) lw 3 lc "#9a031e" dt 3
```

The output plot is given in the following page.

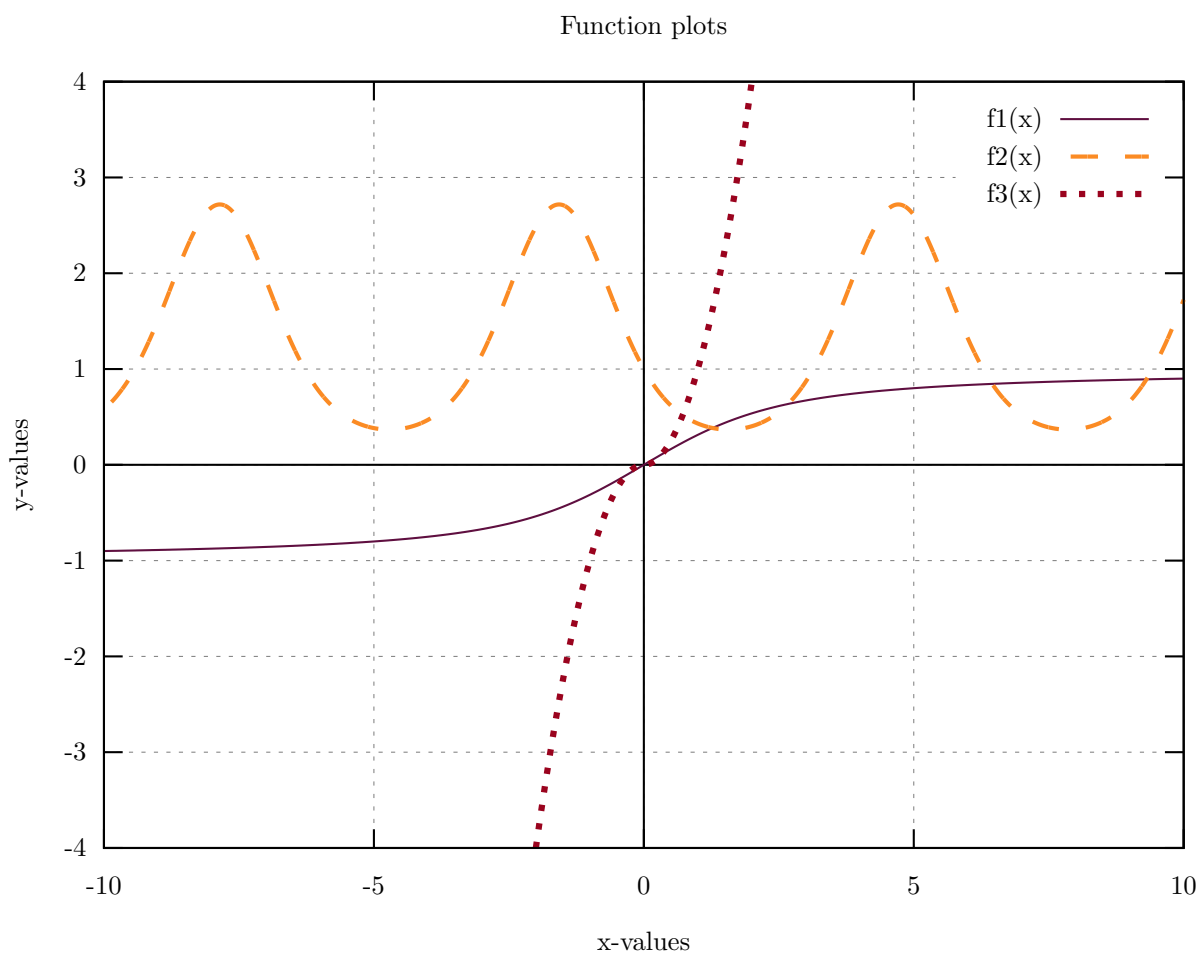


Figure 1: $f_1(x) = \coth(x) - \frac{1}{x}$, $f_2(x) = e^{-\sin(x)}$, $f_3(x) = x|x|$

Question 2

Plot the following functions for $0 \leq x \leq \pi$ on the same graph using GNUPLOT. Set 1000 samples. Mark each curve with different line style and line widths. Label x-axis and y-axis as 'x-values' and 'y-values' respectively. Set the title of the plot 'Function plots'. Save the plot in a suitable format.

$$f_1(x) = \sin^2(\sqrt{x}), \quad f_2(x) = e^{-x/2}, \quad f_3(x) = 12.34x^{5/3}$$

Sol: The GNU-Plot code for the solution of this problem is given below.

```
1 reset
2 set terminal svg background "white"
3 set output "Lab-files/Output/1/Q_xx_ou.svg"
4
5 set title "Function plots"
6 set grid
7 set xzeroaxis linestyle 1 lc "black"
8 set yzeroaxis linestyle 1 lc "black"
9 set xrange [0:pi]
10 set yrange [0:8]
11 set xtics 0.5
12 set ytics 1
13 set xlabel "x-values"
14 set ylabel "y-values"
15
16 f1(x)=sin(sqrt(x))*2
17 f2(x)=exp(-x/2)
18 f3(x)=12.34*(x**(5/3))
19
20 set samples 1000
21 plot f1(x) lw 1 dt 1 lc "#780000", f2(x) lw 2 dt 2 lc "#c1121f", f3(x) lw 3 dt 3
    lc "#003049"
```

The output plot is given in the following page.

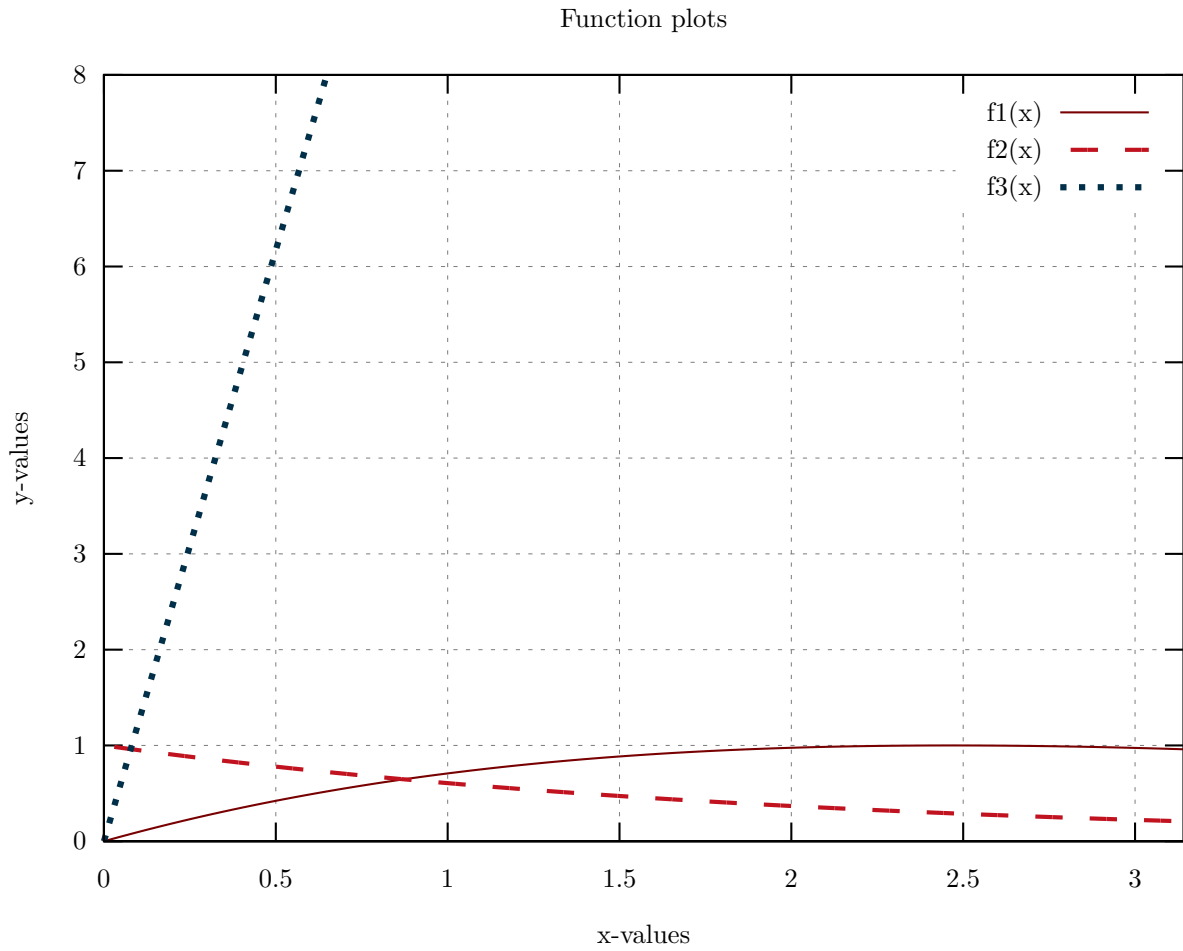


Figure 2: $f_1(x) = \sin^2(\sqrt{x})$, $f_2(x) = e^{-x/2}$, $f_3(x) = 12.34x^{5/3}$

Question 3

Define three functions in the given range:

$$\begin{aligned} f_1(x) &= \ln\left(\frac{1}{x}\right), & x > 0.1, \\ f_2(x) &= 3x^2 \sin x, & -1 < x < 0.1, \\ f_3(x) &= \cos 3x, & x < -1. \end{aligned}$$

Set the range of x values from -3 to $+3$ and the range of y values from -2.5 to $+2.5$. Show both the axes on the plot. Set y label "Piecewise continuous function". Set ticks of the y axis at an interval of 0.5 . Plot the three functions and store the plot in a file and print it.

Sol: The GNU-Plot code for the solution of this problem is given below.

```
1 reset
2 set terminal svg background "white"
3 set output "Lab-files/Output/1/Q_xiv_ou.svg"
4
5 set grid
6
7 set xzeroaxis linestyle 1 lc "black"
8 set yzeroaxis linestyle 1 lc "black"
9
10 set ytics 0.5
11
12 set xrange [-3:3]
13 set yrange [-2.5:2.5]
14
15 set ylabel "Piecewise continuous function"
16
17
18 f1(x)=(x>0.1)?(log(1/x)):NaN #Not as Number
19 f2(x)=(x<0.1 && x>-1)?(3*(x**2)*sin(x)):NaN
20 f3(x)=(x<-1)?(cos(3*x)):NaN
21
22 plot f1(x) lc "#9b2226" lw 2 , f2(x) lc "#bb3e03" lw 2 , f3(x) lc "#ee9b00" lw 2
```

The output plot is given in the figure below.

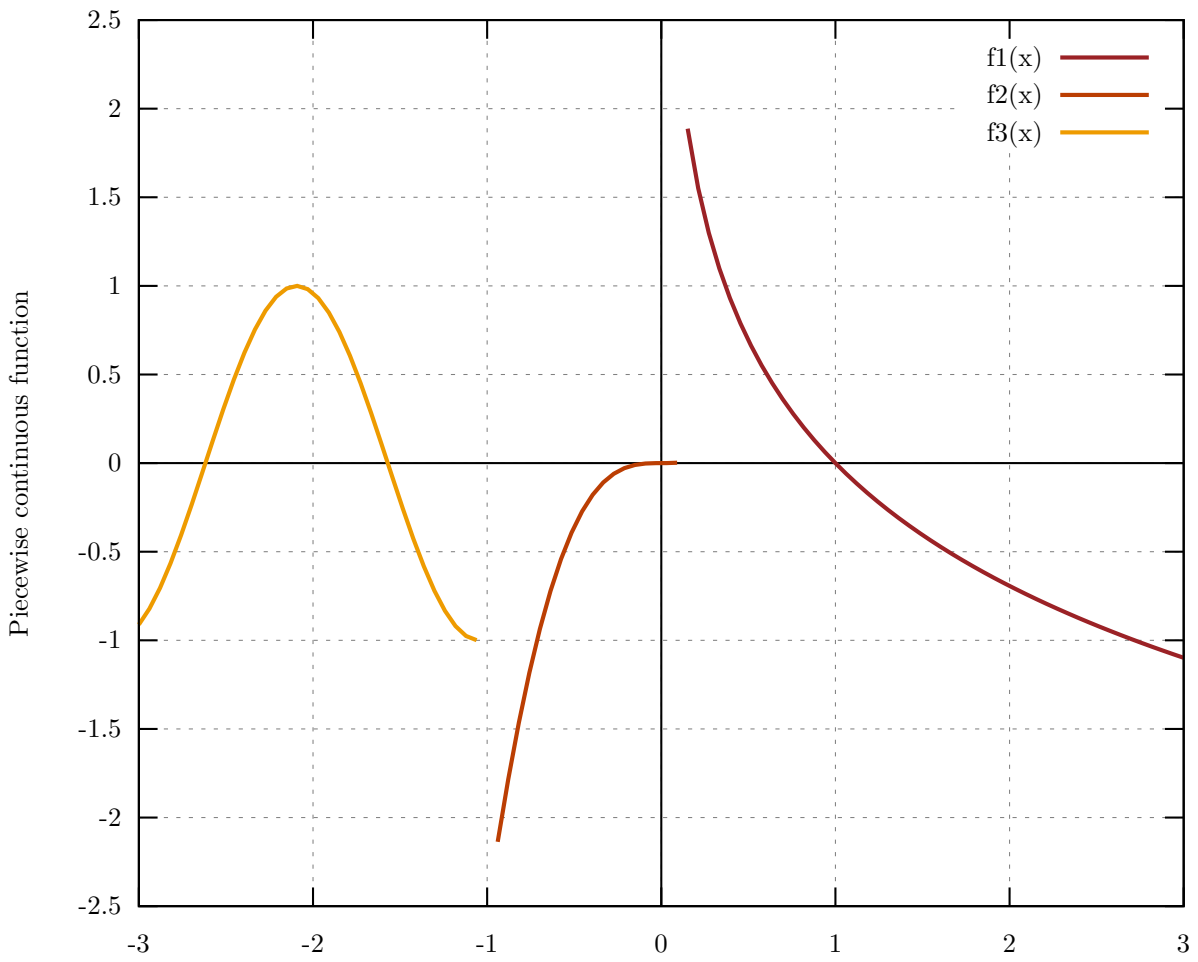


Figure 3: Plot of the piecewise continuous functions $f_1(x)$, $f_2(x)$, and $f_3(x)$

Question 4

Plot

$$2x \sin 2x, \quad \frac{1}{2}x \sinh\left(\frac{x}{2}\right)$$

on the same graph. The line width of the two curves to be different. Set the range of x values from $-\pi$ to $+\pi$ and the range of y values from -5 to 5 . Show both the axes on the plot. Ticks of x axis will be at interval of 0.5 . The label of x and y axis to be "x values" and "y values" respectively. Store the plot in a file and print it.

Sol: The GNU-Plot code for the solution of this problem is given below.

```
1 reset
2 set terminal svg background "white"
3 set output "Lab-files/Output/1/Q_xvi_ou.svg"
4
5 set grid
6 set xzeroaxis linestyle 1 lc "black"
7 set yzeroaxis linestyle 1 lc "black"
8 set xrange [-pi:pi]
9 set yrange [-5:5]
10 set xtics 0.5
11 set xlabel "x values"
12 set ylabel "y values"
13
14 f1(x)=2*x*sin(2*x)
15 f2(x)=(x/2)*sinh(x/2)
16
17 plot f1(x) lc "#9a031e" lw 2, f2(x) lc "#0f4c5c" lw 3
```

The output plot is given in the figure below.

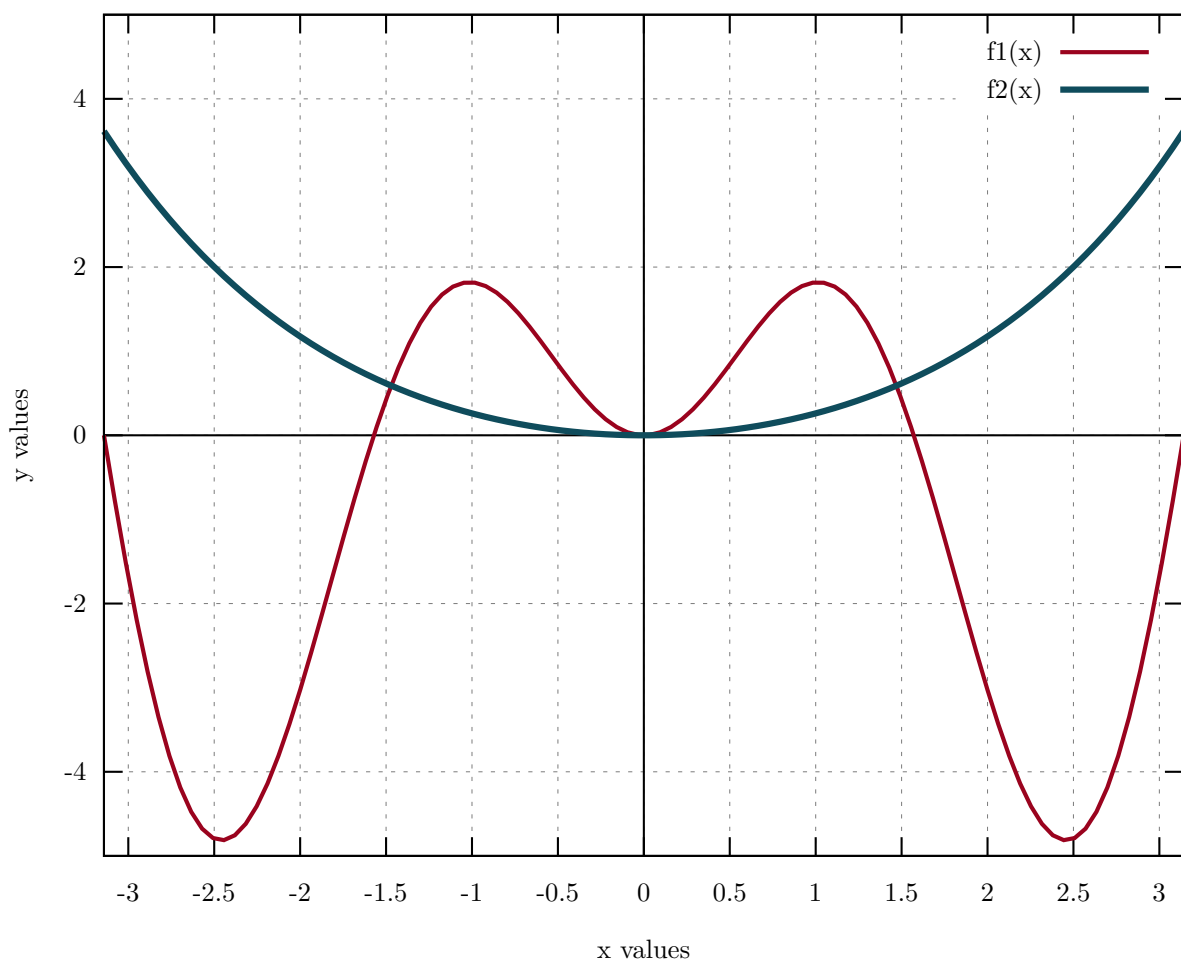


Figure 4: $2x \sin 2x$, $\frac{1}{2}x \sinh\left(\frac{x}{2}\right)$

Question 5

Problem Set 1 (iii). Plot $\sin(x)$ and $2\sin(-x)$ simultaneously on the same plot within the range -3π to 3π . Use the system-defined value of 3π as `3*pi`. Label the x -axis as “x values” and the y -axis as “standing wave”. Save the output plot in a file.

Sol: The GNU-Plot code for the solution of this problem is given below.

```
1 reset
2 set terminal svg background "white"
3 set output "Lab-files/Output/1/Q_iii_ou.svg"
4
5 set grid
6 set xzeroaxis linetype 1 lc "black"
7 set yzeroaxis linetype 1 lc "black"
8 set xrange [-3*pi:3*pi]
9 set yrange [-3:3]
10 set xlabel "x values"
11 set ylabel "standing wave"
12
13 plot sin(x) lc "orchid4" lw 2, 2*sin(-x) lw 2
```

The output plot is given in the figure below.

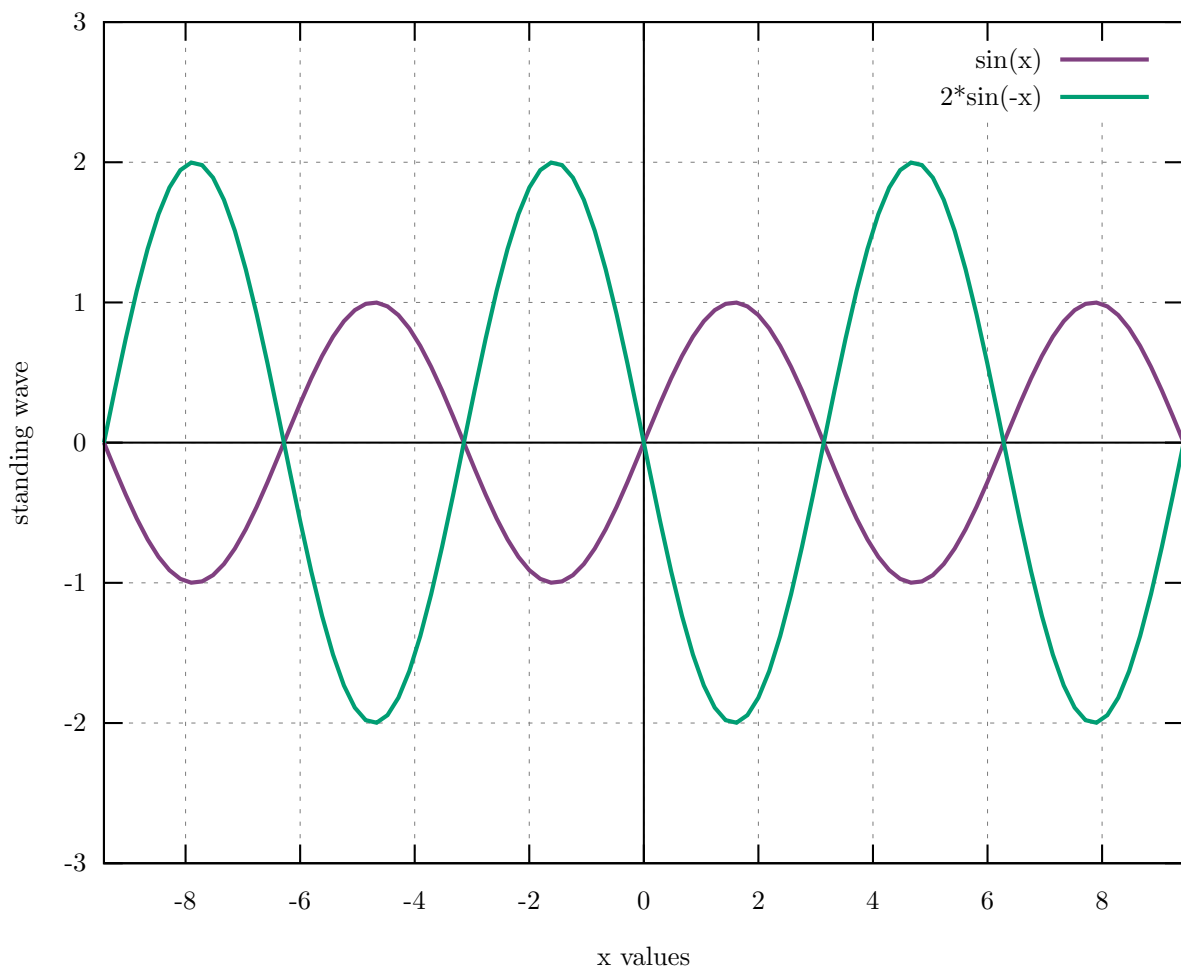


Figure 5: The functions $\sin(x)$ and $2\sin(-x)$ plotted over $-3\pi \leq x \leq 3\pi$.

Question 6

Problem Set 1 (xiii). Define the following piecewise function:

$$f(x) = \begin{cases} x \cos x, & x < 0, \\ x^2 + 3x, & 0 \leq x \leq 1, \\ \ln x + 4, & x > 1. \end{cases}$$

Plot it for $-3 \leq x \leq 3$ and $-2 \leq y \leq 5$. Show both axes and set the title to “Continuous function without derivative at all points”. Save the plot in a file.

Sol: The GNU-Plot code for the solution of this problem is given below.

```
1 reset
2 set terminal svg background "white"
3 set output "Lab-files/Output/1/Q_xiii_ou.svg"
4
5 set title "Continuous function without derivative at all points"
6 set grid
7 set xzeroaxis linestyle 1 lc "black"
8 set yzeroaxis linestyle 1 lc "black"
9 set xrange [-3:3]
10 set yrange [-2:5]
11
12 f(x)=(x<0)?(x*cos(x)):(x>=0 && x<=1)?(x**2 + 3*x):(x>1)?(log(x)+4):NaN
13
14 plot f(x) lc "#ee6c4d" lw 2
```

The output plot is given in the figure below.

Continuous function without derivative at all points

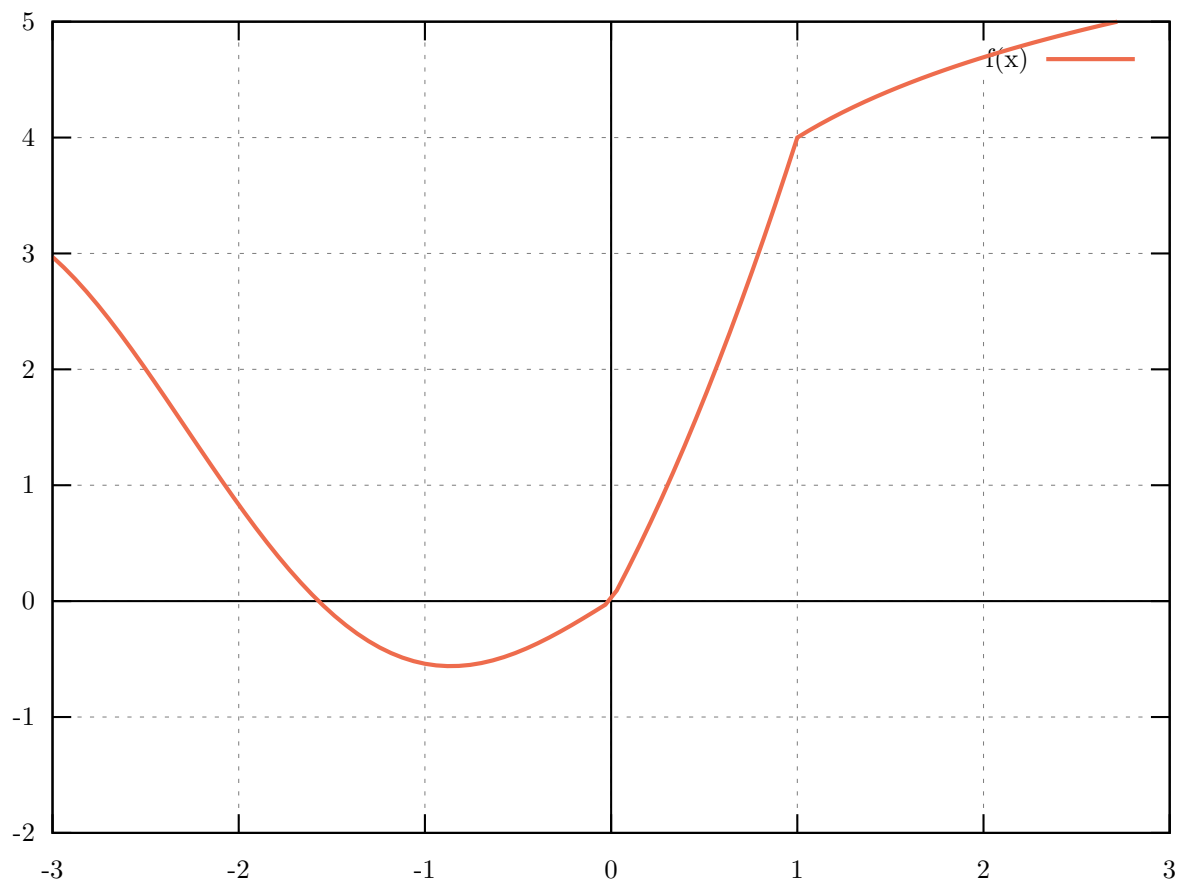


Figure 6: A continuous piecewise function with changes in derivative.

Question 7

Problem Set 1 (xii). Plot $\frac{\sin x}{x}$, $\frac{\sinh x}{x}$, and $y = 1$ on the same graph. Set the x -range from $-\pi$ to $+\pi$ and the y -range from 0 to 5. Show the y -axis, set the title to “Plot of different functions of x ”, and label the axes “ x values” and “ y values”. Use different line widths for all three curves and save the plot in a file.

Sol: The GNU-Plot code for the solution of this problem is given below.

```
1 reset
2 set terminal svg background "white"
3 set output "Lab-files/Output/1/Q_xii_ou.svg"
4
5 set grid
6 set title "Plot of different functions of x"
7 set xrange [-pi:pi]
8 set yrange [0:5]
9 set xtics 0.5
10 set ytics 1
11 set xlabel "x values"
12 set ylabel "y values"
13 set xzeroaxis linestyle 1 lc "black"
14 set yzeroaxis linestyle 1 lc "black"
15
16 y1(x)=(x==0)?1:sin(x)/x
17 y2(x)=(x==0)?1:sinh(x)/x
18 y3(x)=1
19
20 plot y1(x) lc "#6F1D1B" lw 1, y2(x) lc "#fb8500" lw 2, y3(x) lc "#f94144" lw 3
```

The output plot is given in the figure below.

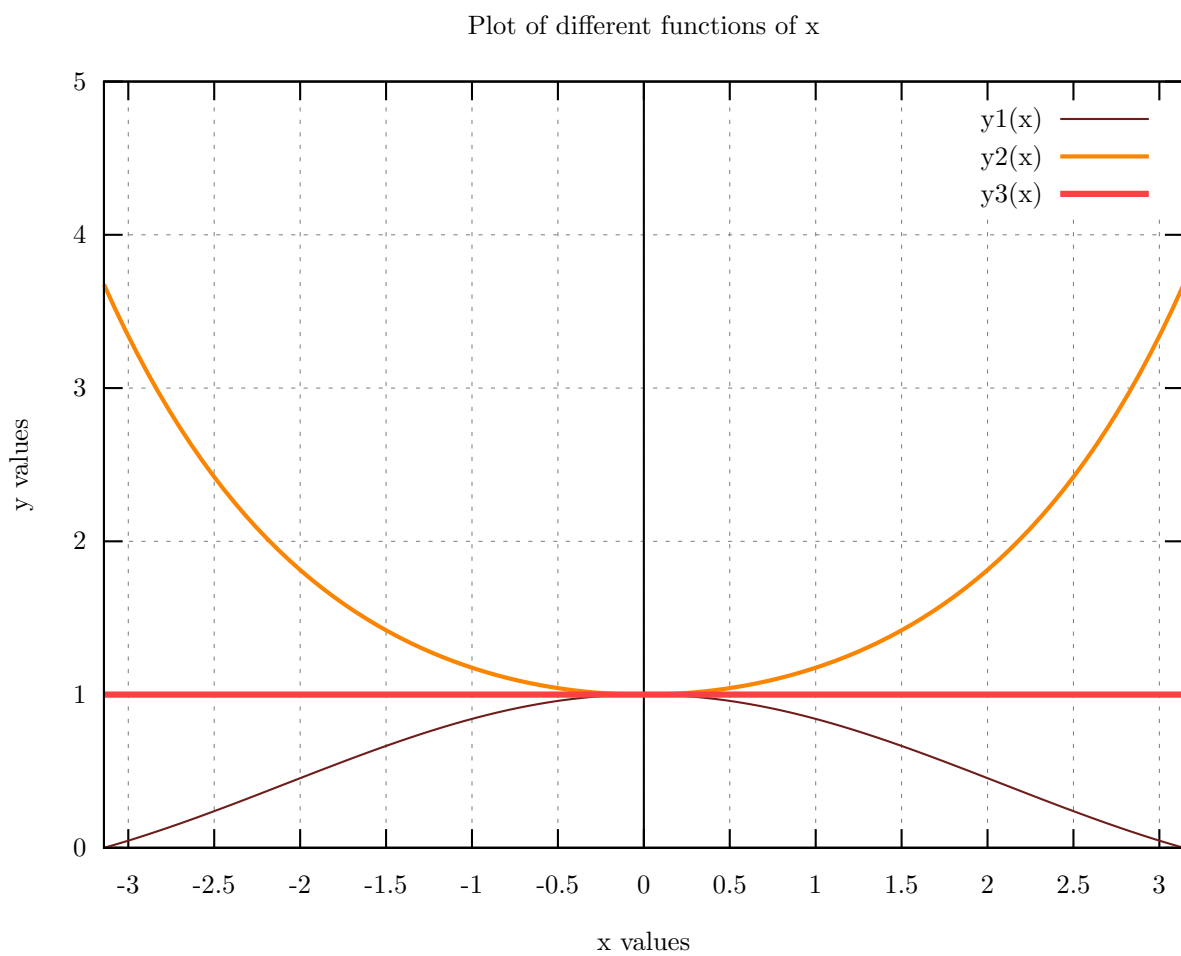


Figure 7: The functions $\frac{\sin x}{x}$, $\frac{\sinh x}{x}$, and $y = 1$.